Maths: Multiplying Fractions

Skill: Written method and checking | Warm-up 5, drill 10, paper 20 min

Writing target

- Do not just show numbers. For every step, write one short sentence saying what you did and why, using the link words below.
- Link words: so / because / which means / therefore / this works because / first … then …

The method (use it every time)

- 1. Turn mixed numbers into improper fractions.
- 2. Cancel across first: any top and any bottom that share a factor can be divided down.
- 3. Multiply the tops. Multiply the bottoms.
- 4. Simplify, and turn back into a mixed number if the answer is top-heavy.
- 5. Check: round everything to whole numbers and estimate. Is your answer close?

A. No calculator. Show every step.

(a) 2/3 × 3/4 = (b) 5/6 × 9/10 = (c) 3/8 × 4/9 = (d) 3/5 of 40 =

B. Mixed numbers.

(e) 1½ × 2 2/3 = (f) 2 1/4 × 1 1/3 =

C. Estimate first, then calculate.

3 4/5 × 2 1/6. My estimate: _____ Exact answer: _____ Close enough? Yes / No

D. Spot the error.

A pupil writes 1½ × 1½ = 1¼ (they did 1×1 = 1 and ½×½ = ¼). What went wrong, and what is the correct answer? Write two sentences.

E. Explain it in words.

Explain why 2/3 × 3/4 is smaller than both 2/3 and 3/4. Then explain why we can cancel before multiplying. Use *because* or *which means* in each answer.

F. Worked solution in sentences (writing practice).

A tank holds 2½ litres. 3/5 of it is used, then 1/3 of what is left is poured out. How much is left? Write the solution as **sentences**, one per step, each saying what you did and why. Start: *First, I turned 2½ into an improper fraction because…*

Self-check: [ ] Mixed numbers converted first [ ] Cancelled before multiplying [ ] Simplified [ ] Estimate matches [ ] Every step has a because

Mark your writing: circle the level you reached.

Level	What it looks like
Level 1	Numbers only. Answer may be right but the working is not explained.
Level 2	Words say what was done at each step.
Level 3	Words say what was done AND why each step is allowed.